Surrogate-Assisted Double Machine Learning for Discrete Outcomes

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Abstract

Applying partialling-out directly to a discrete label targets the label's numeric scale, not a cumulative-link or log-mean coefficient. Surrogate-Assisted Double Machine Learning (SUDO) restores the target scale before partialling-out. A fitted distribution supplies a scalar index and conditional CDF; randomization within the observed CDF jump then produces a continuous surrogate whose conditional mean equals the true index when the distribution is correct. Standard Frisch-Waugh-Lovell partialling-out can then recover the treatment coefficient. Proper multiple-imputation draws provide inference for parametric fits, while flexible learners require a full-pipeline bootstrap. Simulations establish nominal coverage for the validated binary and ordinal designs and show the central limitation: double-machine-learning orthogonality does not protect against error in the imputation distribution. Scalar recalibration removes regularization bias from a coefficientless additive learner in three designs, but fails at a high-dimensional boundary. SUDO agrees with correct direct and tailored binary estimators; its practical value is a common route to latent-scale estimation, especially for ordinal outcomes. A wine-quality analysis illustrates that use. An oracle experiment validates count completion, but no fitted count learner is yet supported.

1 Introduction

Many important outcomes are discrete: relapse is binary, survey responses are ordinal, and trips are counts. Double Machine Learning (DML, Chernozhukov et al. 2018) handles a continuous partially linear outcome by removing flexibly estimated covariate effects and regressing one residual on another. The same calculation can be applied to a discrete label, but it then targets a contrast in $E[Y \mid D, X]$. It does not recover a cumulative-link coefficient or a log conditional-mean effect. The obstacle is the outcome scale.

SUDO changes the outcome before applying DML. It draws a continuous value inside the fitted CDF jump for the observed category and adds the fitted scalar index. If the conditional distribution is correct, the completed outcome has that index as its conditional mean. The treatment coefficient is therefore available through ordinary partialling-out.

This construction turns the surrogate-variable method of D. Liu and Zhang (2018) from a diagnostic into an estimator. Cheng, Wang, and Zhang (2021) extend surrogate residuals to unordered discrete-choice models with vector utilities, and Greenwell et al. (2018) implement binary and ordinal diagnostics in the `sure` package. SUDO instead focuses on scalar-index distributions. The learner supplies an index and CDF; SUDO supplies completion, partialling-out, and uncertainty propagation.

The common interface is useful, but it is not an efficiency claim. Tailored orthogonal scores already exist for logistic binary models (M. Liu, Zhang, and Zhou 2021) and Poisson log-mean effects (Drukker and Liu 2022). SUDO agrees with those estimators when their models are correct. Its clearest use is ordinal regression, where the same construction targets a latent proportional-odds coefficient without deriving a new score. The simulations also establish the method's main boundary: DML protects the adjustment regressions, not the distribution that creates the surrogate. New imputation learners therefore require direct bias and coverage validation.

2 From a discrete outcome to the target scale

Let D be a treatment, $X \in \mathbb{R}^p$ covariates, and Y a discrete outcome with conditional CDF $G_0(y \mid D, X)$. The true distribution has a scalar additive index

$$\eta_0(D, X) = \theta_0 D + g_0(X). \tag{1}$$

For binary and ordinal outcomes this is a latent-utility index: binary $Y = \mathbf{1}\{Z > 0\}$, or ordinal, $Y = j \iff \kappa_{j-1} < Z \leq \kappa_j$ for thresholds $-\infty = \kappa_0 < \kappa_1 < \cdots < \kappa_{J-1} < \kappa_J = \infty$. The latent utility follows a partially linear model

$$Z = \theta_0 D + g_0(X) + \varepsilon, \qquad \varepsilon \mid D, X \sim F \tag{2}$$

with g_0 an arbitrary nuisance function and F a known error distribution (logistic for proportional odds, extreme-value for complementary log-log, and so on). The target θ_0 is the treatment effect on this latent scale, with g_0 left nonparametric. It is not a probability-scale average treatment effect. Under a logistic link, $\exp(\theta_0)$ is a proportional-odds ratio. For Poisson or negative-binomial counts we use $E[Y \mid D, X] = \exp\{\eta_0(D, X)\}$, making θ_0 a log conditional-mean effect and $\exp(\theta_0)$ a conditional rate ratio.

This notation matches the surrogate literature up to sign convention. D. Liu and Zhang (2018) write the ordinal latent variable as $Z = -f(X, \beta) + \varepsilon$, draw the surrogate from $S \mid (Y, X) \sim Z \mid (Y, X)$, and define the surrogate residual $R^S = S - E_0(S \mid X)$. Here $\eta = -f$, treatment is included in the index, and the reference error is centered, so the corresponding correct-model residual is $R^S = S - \eta(D, X)$. SUDO uses the surrogate S as an outcome for estimation rather than using R^S only for diagnostics. The vector surrogate of Cheng, Wang, and Zhang (2021) is broader, but is not required for the scalar estimands studied here.

Identification requires four conditions.

1. *Selection on observables.* No unobserved confounder acts beyond X. Together with the conditional location-family statement in Equation 2, this assumption requires the latent error to have the same normalized law across treatment values given X.
2. *A constant index-scale effect.* The partially linear form in Equation 1 makes θ_0 a single number rather than a function of X.
3. *A correctly specified discrete conditional distribution.* Completion uses the entire fitted CDF jump, including the link and thresholds for cumulative models or the family and dispersion for counts.
4. *Overlap.* Treatment retains support after adjustment, so its residual is not annihilated.

The first and fourth are standard causal conditions. The second and third are modeling choices shared with any structured distributional analysis.

2.1 Randomized completion

FWL estimates a treatment coefficient by regressing an outcome residual on a treatment residual after removing X from both. DML combines that identity with cross-fitting and flexible nuisance regressions (Chernozhukov et al. 2018). SUDO changes only the outcome supplied to this calculation.

Fit a structured distributional model of Y on (D, X). It must return the scalar index $\hat{\eta}_i$, the fitted CDF immediately below and at the observed outcome, and any global distribution parameters. Draw

$$V_i \mid Y_i, D_i, X_i \sim \mathrm{Unif}\{\hat{G}(Y_i - 1 \mid D_i, X_i), \hat{G}(Y_i \mid D_i, X_i)\}, \qquad S_i = \hat{\eta}_i + H^{-1}(V_i), \tag{3}$$

where H is a continuous mean-zero reference law. For binary and ordinal cumulative links, we set H to the centered link-error law. This gives the truncated latent-variable surrogate of D. Liu and Zhang (2018), expressed through its CDF jump, up to the corresponding intercept centering. For counts we use standard normal H, following the randomized-quantile-residual convention of Dunn and Smyth (1996). The reference law changes the completion variance but not the target.

For the normal count reference, the completion has an analytic Rao-Blackwell form. With $a = \hat{G}(Y - 1 \mid D, X)$, $b = \hat{G}(Y \mid D, X)$, and standard-normal density ϕ,

$$E\{\Phi^{-1}(V) \mid Y, D, X\} = \frac{\phi\{\Phi^{-1}(a)\} - \phi\{\Phi^{-1}(b)\}}{b - a},$$

with the continuous limiting value for a zero-width numerical jump. The count stages report this analytic completion beside the $B = 25$ randomized completion.

Proposition 1 (randomized-CDF completion). *If $G = G_0$ and $\eta = \eta_0$, and if H is continuous with $E\{H^{-1}(V)\} = 0$ for $V \sim \mathrm{Unif}(0, 1)$, then the completion in Equation 3 satisfies $E[S \mid D, X] = \eta_0(D, X)$. Consequently,*

$$S = \theta_0 D + g_0(X) + \eta, \qquad E[\eta \mid D, X] = 0$$

and population FWL recovers θ_0.

Proof. Conditional on (D, X), each discrete outcome selects one disjoint CDF jump with probability equal to that jump's width. Uniform randomization inside the selected jump therefore makes $V \mid D, X$ uniform on $(0, 1)$. The reference quantile has conditional mean zero, leaving $\eta_0(D, X)$. □

For cumulative links, this is the conditioning identity behind the surrogate-variable results of D. Liu and Zhang (2018) and Cheng, Wang, and Zhang (2021). The estimation step is to use S, rather than only its diagnostic residual, as the outcome in FWL or DML.

With an estimated distribution, the equality is asymptotic under consistency and approximate otherwise. The imputation model is not protected by DML orthogonality.

The following base R example compares the fitted logistic coefficient, FWL on the unobserved utility, and FWL on one surrogate completion.

```r
set.seed(1)
n <- 40000
theta <- 1.5
X <- rnorm(n)
D <- rbinom(n, 1, plogis(0.8 * X))
Z <- theta * D + X + rlogis(n)
Y <- as.integer(Z > 0)

imputation_fit <- glm(Y ~ D + X, family = binomial())
index_hat <- predict(imputation_fit, type = "link")
cdf_at_zero <- plogis(-index_hat)

lower <- ifelse(Y == 1, cdf_at_zero, 0)
upper <- ifelse(Y == 1, 1, cdf_at_zero)
uniform_draw <- runif(n, min = lower, max = upper)
uniform_draw <- pmin(pmax(uniform_draw, 1e-9), 1 - 1e-9)
surrogate <- index_hat + qlogis(uniform_draw)

treatment_residual <- resid(lm(D ~ X))
fwl_coefficient <- function(outcome) {
  sum(treatment_residual * resid(lm(outcome ~ X))) /
    sum(treatment_residual^2)
}
round(c(
  glm = unname(coef(imputation_fit)["D"]),
  oracle = fwl_coefficient(Z),
  surrogate = fwl_coefficient(surrogate)
), 3)
```

```
      glm    oracle surrogate
    1.497     1.496     1.508
```

All three estimates are close to $\theta_0 = 1.5$. Using only the binary labels, the surrogate completion recovers the coefficient that FWL would estimate if the latent utility were observed. Replacing the binary index with an ordinal cumulative-link index and truncating to category-specific intervals gives the ordinal construction. The remainder of the paper adds two elements to this basic estimator: machine-learning partialling and inference that accounts for the uncertainty hidden by a single completion.

3 Estimation and inference

Cross-fit the two adjustment nuisances $\hat{m}(X) \approx E[D \mid X]$ and $\hat{\ell}(X) \approx E[S \mid X]$, and form the residual-on-residual estimate

$$\hat{\theta} = \frac{\sum_i (S_i - \hat{\ell}(X_i))(D_i - \hat{m}(X_i))}{\sum_i (D_i - \hat{m}(X_i))^2} \tag{4}$$

The $\hat{\ell}$ regression uses X only: the index carried D's signal into S, and partialling on X alone isolates θ_0. The score $\varphi = (S - \ell(X) - \theta(D - m(X)))(D - m(X))$ is Neyman-orthogonal in (ℓ, m): its derivatives in both nuisance directions vanish by the tower property, so first-order error in $\hat{\ell}$ or $\hat{m}$ does not bias $\hat{\theta}$. This protection covers the two adjustment nuisances, not the imputation model that produced the surrogate.

One completion understates uncertainty. We therefore draw B surrogates and pool them with Rubin's rules (Rubin 1987): the estimate is $\bar{\theta} = B^{-1} \sum_b \hat{\theta}^{(b)}$ and the total variance is $T = \bar{W} + (1 + 1/B)B_*$, with a Barnard-Rubin small-sample t reference (Barnard and Rubin 1999). Reusing one fitted $\hat{\beta}$ across all completions leaves its shared estimation error invisible to the between-draw variance. A proper draw also redraws the imputation model's parameters, $\beta^{(b)} \sim N(\hat{\beta}, \widehat{\mathrm{Cov}}(\hat{\beta}))$ (thresholds included for ordinal outcomes), before each completion. This is the large-sample normal form of a proper posterior-predictive draw. It accounts for sampling noise, surrogate randomness, and imputation-model uncertainty. Table 1 shows why all three matter.

For ordinal outcomes, thresholds are parameters too and must be perturbed jointly with the coefficients. Surrogate-residual mean and variance checks provide a diagnostic for the assumed link and mean structure (D. Liu and Zhang 2018), but they do not replace target-level validation.

3.1 What completion contributes

Let $R_D = D - m_0(X)$ and let a fitted distribution learner return the pointwise index $\hat{\eta}(D, X)$ without requiring a treatment coefficient. Write $\mu_Y = E\{H^{-1}(V) \mid Y, D, X, \hat{G}\}$ for the conditional mean reference correction. The randomized estimator converges to the same completion mean as its Rao-Blackwellized form.

Proposition 2 (index projection plus correction). *With the population treatment nuisance, the population completion target is*

$$\theta_{\mathrm{SUDO}} = \frac{E\{R_D \hat{\eta}(D, X)\}}{E(R_D^2)} + \frac{E(R_D \mu_Y)}{E(R_D^2)}. \tag{5}$$

Proof. Substitute $S = \hat{\eta}(D, X) + \mu_Y$ into the population FWL numerator and separate its two terms. If $\hat{\eta}(D, X) = \hat{\alpha} D + \hat{f}(X)$, then $E\{R_D \hat{f}(X)\} = 0$ and $E(R_D D) = E(R_D^2)$, so the first term reduces to $\hat{\alpha}$. □

The first term projects the fitted index onto residualized treatment; the second uses the observed outcome to correct that projection. For an additive fit, the first term is its treatment coefficient. The correction can offset treatment shrinkage, transmit covariate-direction error, or vanish. Under a binary logit model, it is an entropy-weighted residual moment.

3.2 The unprotected imputation model

Orthogonality protects $\hat{\theta}$ from first-order error in the adjustment nuisances. It does not protect the imputation model: its error becomes part of the completed outcome supplied to DML.

Here, a *black-box imputation model* means a flexible distributional learner that supplies a pointwise scalar index and conditional CDF. It need not parameterize or expose a treatment coefficient. The count validation isolates D from the joint X interaction block. An arbitrary probability vector without a scalar index on the target scale remains outside scope. Flexible imputation models are cross-fit so each supplied index is an out-of-fold prediction.

Conditional on the fitted model, $\hat{\eta}$ is fixed while $Y \mid D, X$ is generated by the truth. Since S is $\hat{\eta}$ plus truncated noise,

$$E[S \mid D, X] = \psi(\hat{\eta}, \eta), \quad \psi(v, e) = v + F(e)\, \mu_+(v) + (1 - F(e))\, \mu_-(v) \tag{6}$$

with $\mu_\pm$ the truncated-noise mean corrections. The correctly centered draw is exact, $\psi(e, e) = e$, the correct-model surrogate identity in D. Liu and Zhang (2018) and Cheng, Wang, and Zhang (2021), but index error passes through with a factor strictly below one,

$$c(e) = \frac{\partial \psi}{\partial v}\Big|_{v=e} = 1 - f(-e)\,(\mu_+(e) - \mu_-(e)), \qquad f = F' \tag{7}$$

Read c as the fraction of index error that survives the truncation. With no truncation the surrogate would be index plus noise and its mean would move one-for-one with $\hat{\eta}$, giving $c = 1$; the observed category drags it back, and c is what is left.

Under a logistic link, c rises from $1 - \log 2 = 0.31$ at $e = 0$ toward one in the tails (Figure 1). In fact,

$$c(e) = 1 - H(F(e)), \qquad H(p) = -p \log p - (1-p)\log(1-p), \tag{8}$$

so self-correction equals outcome entropy. An uncertain label sharply constrains the latent utility and corrects more of the fitted-index error. A nearly deterministic label adds little information and corrects almost none.

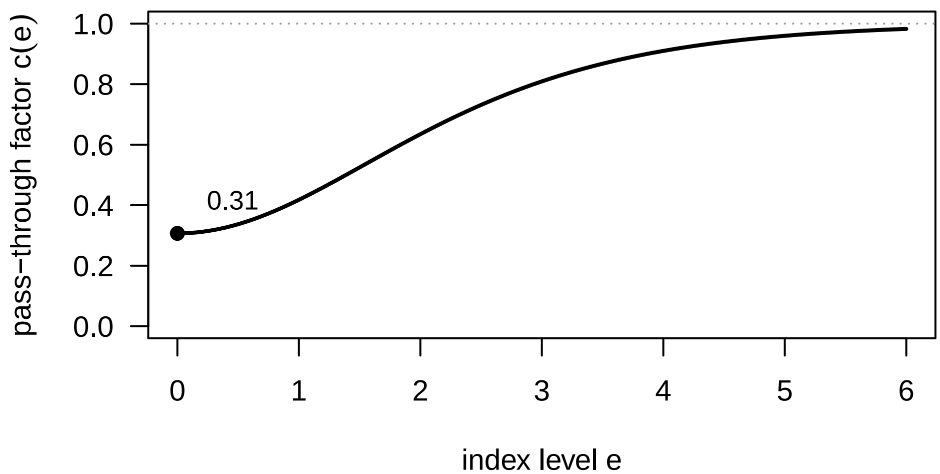

Figure 1: Fraction of fitted-index error that survives completion under a logistic link. An informative label supplies the strongest correction near the center; the correction fades as the label becomes nearly deterministic.

Fitted-index error therefore enters the SUDO target at first order. If the effective treatment component is displaced by δ_D, the leading bias is approximately $\bar{c}_1\delta_D$, where $\bar{c}_1$ averages $c(e)$ over the treatment-relevant index distribution. Error in the covariate component can also survive because completion is nonlinear in the fitted index. Partialling-out does not erase either channel. Predictive fit alone is not a sufficient validation criterion; the treatment target must be checked directly.

4 Flexible distribution learners

A parametric model supplies an index and covariance matrix, making proper draws straightforward. A flexible learner may supply neither a treatment coefficient nor a usable joint covariance. SUDO asks it only for pointwise index and CDF predictions, then handles structure and inference separately.

For the validated binary experiments, we restrict the imputation index to $\hat{\alpha}D + \hat{f}(X)$. A GAM, MARS bases built from X only, and neural-network backfitting all implement this structure. The restriction prevents treatment-by-covariate interactions, but it does not guarantee a good fit or a small target bias. Two other partially linear learners failed the bias criterion, including one tuned by predictive risk.

When a learner has no usable covariance, a full-pipeline bootstrap resamples the observations and reruns the entire procedure, including the imputation learner, folds, completions, and adjustment nuisances. A within-fold refit misses important sources of variation. In the validated binary designs, normal intervals based on the full-pipeline bootstrap standard deviation attain nominal coverage.

The resulting workflow is deliberately strict: impose a scalar-index structure suited to the estimand, tune on target-level simulation bias, and check coverage before using a learner for inference.

4.1 Scale-calibrated SUDO

Regularization can preserve the shape of an index while compressing its location and scale. Scalar recalibration corrects that error before completion. Let a learner supply an out-of-fold logit index $\hat{\eta}(D, X)$. On the training observations for a held-out fold, fit

$$P(Y = 1 \mid \hat{\eta}) = \Lambda(a + b\hat{\eta}) \tag{9}$$

and apply $\tilde{\eta} = a + b\hat{\eta}$ to the held-out predictions. The estimator does not read a treatment coefficient from the learner, and neither treatment nor its residual enters the calibration regression.

For binary logit, write $p = \Lambda(\tilde{\eta})$ and $h(p) = -p \log p - (1 - p) \log(1 - p)$. The Rao-Blackwell surrogate is

$$\tilde{S} = \tilde{\eta} + \frac{h(p)\{Y - p\}}{p(1 - p)}. \tag{10}$$

It equals the conditional mean of a truncated-logistic surrogate and removes completion Monte Carlo noise. Randomized completion remains available by drawing the logistic reference quantile inside the fitted Bernoulli CDF jump.

Algorithm 1: Cross-fitted scale-calibrated SUDO.

1. Split the data into K folds and fix the tuning rule.
2. For each held-out fold, fit the distribution learner and treatment nuisance $m(X) = E(D \mid X)$ on the other folds. Predict the scalar index in the training and held-out observations.
3. Fit Equation 9 on the training predictions and recalibrate the held-out index.
4. Use Equation 10 to construct training and held-out surrogates. Fit $\ell(X) = E(\tilde{S} \mid X)$ on the training observations and predict it in the held-out fold.
5. Stack held-out predictions and compute Equation 4 with $S = \tilde{S}$. For randomized completion, repeat the completion and partialling steps, then average the treatment estimates.
6. With a flexible learner, bootstrap the observations and rerun all five steps. Form a normal interval from the bootstrap standard deviation, $\hat{\theta} \pm 1.96\widehat{\mathrm{sd}}_*$.

The reason for the calibration is exact in a useful special case. If, on a new population, $\eta_0(D, X) = a_0 + b_0\hat{\eta}(D, X)$ and scalar recalibration consistently estimates (a_0, b_0), then Equation 10 gives $E(\tilde{S} \mid D, X) = \eta_0(D, X)$. Population FWL therefore recovers θ_0. With a remaining error $r(D, X)$, bias depends on its projection onto $D - m_0(X)$ after completion. Scalar calibration does not control an arbitrary r and is not a substitute for target-level validation.

For sensitivity analysis, an affine fluctuation can add a treatment-residual direction to Equation 9. That version partly targets the treatment effect before completion. Scale-only calibration more cleanly isolates what the surrogate and FWL steps contribute.

5 Simulation evidence

The simulations draw independent standard-normal covariates, assign treatment from those covariates, and coarsen a partially linear latent utility:

$$\begin{gathered} X_1, \dots, X_p \overset{\text{iid}}{\sim} N(0, 1), \qquad D \mid X \sim \text{Bernoulli}(\Lambda(\pi(X))), \qquad Z = \theta_0 D + g_0(X) + \varepsilon, \\ \varepsilon \sim F, \qquad Y = j \iff \kappa_{j-1} < Z \le \kappa_j, \end{gathered} \tag{11}$$

One correctly specified covariate isolates the inference calculation. The five-covariate machine-learning design uses

$$\pi(X) = X_4 + \cos X_5, \qquad g_0(X) = X_1^2 + \sin X_2 + 0.5X_3,$$

with binary or three-category ordinal outcomes. Table 9 lists the full designs.

Coverage is evaluated against 0.95. Standard deviation (SD) measures run-to-run variation; mean standard error (mean SE) averages the reported uncertainty. Every table is generated from a committed result file.

Proper draws deliver coverage. In the simple binary design, the naive interval covers 0.80 and repeated completion with fixed imputation parameters covers 0.92. Redrawing those parameters raises coverage to 0.95 (Table 1).

Table 1: Inference schemes, binary outcome, $n = 5000$, $\theta_0 = 1.5$, 500 replications. B is the number of surrogate draws.

Scheme	Bias	SD	Mean SE	Coverage
Naive single draw	+0.003	0.083	0.055	0.804
Improper, $B = 25$	+0.004	0.071	0.067	0.924
Proper, $B = 25$	+0.005	0.072	0.076	0.970

Ordinal outcomes. With three categories and machine-learning adjustment, SUDO has bias 0.016 and coverage 0.960 in 200 replications (Table 2). Refitting $E[S \mid X]$ for each completion is necessary; reusing one fit inflated the within-draw variance.

Table 2: Ordinal outcome, $J = 3$, $\theta_0 = 1.0$. Parametric arm 500 replications; corrected spline + ML arm 200, with the outcome nuisance refit per completion.

Configuration	Bias	SD	Mean SE	Coverage
Parametric, $n = 3000$	+0.002	0.077	0.083	0.976
Spline + ML nuisances, $n = 2000$	+0.016	0.104	0.114	0.960

Direct and tailored comparisons. In the correctly specified ordinal design, SUDO and the cumulative-link coefficient agree. Under a cloglog truth, SUDO, the tailored orthogonal score, and the direct cloglog coefficient have indistinguishable bias. In the high-dimensional binary design, SUDO reduces treatment-regularization bias but does not remove it (Table 3). These comparisons establish agreement with correct outcome-specific estimators, not an efficiency advantage.

Table 3: Direct and tailored-score comparisons. The table reports the validated common-sample cells most relevant to the scope claim.

Design	Estimator	Bias	SD
Ordinal, correct	Direct	+0.012	0.085
Ordinal, correct	SUDO	+0.011	0.084
Binary high-dimensional, pen	Tailored orthogonal	+0.029	0.116
Binary high-dimensional, pen	SUDO	-0.105	0.110
Binary cloglog, correct link	SUDO	+0.004	0.081
Binary cloglog, correct link	Tailored orthogonal	+0.004	0.081
Binary cloglog, correct link	Direct	+0.003	0.079

Link misspecification. Assuming a logistic link under Gumbel noise biases the latent-scale effect by +0.77 for binary outcomes and +0.58 for ordinal outcomes, with zero coverage. Bias is below 0.005 under the correct link. Conditional second-moment drift in the surrogate residual flags the poor fit, but is not specific to the link and remains a diagnostic (D. Liu and Zhang 2018).

Partially-linear imputation learners. GAM, MARS, and neural backfitting fit the same $\alpha D + f(X)$ structure. All three pass the prespecified bias criterion, and full-pipeline-bootstrap normal coverage ranges from 0.94 to 0.98 (Table 4). With 50 replications, these differences are too small to rank the learners.

Table 4: Partially-linear imputation learners with full-pipeline bootstrap, $n = 2000$, 50 replications, 50 outer resamples, and 15 surrogate completions per resample.

Learner	θ_0	Bias	SD/Mean SE	Normal coverage	Percentile coverage
GAM	1.5	+0.010	0.948	0.980	0.940
GAM	3.0	+0.043	0.960	0.940	0.880
MARS	1.5	-0.009	0.978	0.960	0.920
MARS	3.0	-0.010	0.945	0.960	0.940
Neural backfit	1.5	-0.005	0.978	0.940	0.940
Neural backfit	3.0	+0.007	0.998	0.980	0.960

Partial linearity alone is not enough. Two other partially linear learners fail the bias screen, including one selected by cross-validated predictive risk. The successful configurations were screened on target-level Monte Carlo bias and evaluated on separate seeds. A data-only selection rule remains open.

Coefficientless additive learner. GAMSEL (Chouldechova and Hastie 2015) fits the binary index with a linear treatment term while allowing each X_j to enter linearly, smoothly, or not at all. SUDO uses only its index predictions. In three $p = 80$ additive designs, full-affine calibrated SUDO passes every prespecified bias, coverage, SD-to-SE, and bootstrap-success gate (Table 5). Ordinary SUDO retains material regularization bias.

Table 5: Coefficientless GAMSEL learner with treatment coefficient 0.75. Full-affine results use 50 fresh datasets and 99 full-pipeline bootstrap resamples.

Design	Ordinary bias	Calibrated bias	Coverage	SD/Mean SE
Randomized	-0.057	-0.012	0.980	0.917
Confounded	-0.076	-0.014	0.940	0.976
High signal	-0.102	-0.009	0.960	1.020

Table 6: Scale-only mechanism screen on separate seeds. Entries are bias over 10 fresh datasets.

Design	Direct read-off	Scale-calibrated SUDO
Randomized	+0.019	+0.014
Confounded	+0.039	+0.009
High signal	+0.009	-0.009

The scale-only ablation separates calibration from treatment targeting. Its biases are $+0.014$, $+0.009$, and -0.009. Under confounding, the coefficient read directly from the recalibrated index has bias $+0.039$; completion and FWL reduce it to $+0.009$ (Table 6). Tilt-only calibration fails under confounding and high signal. Scale-only coverage has not been tested, so the two tables deliberately separate its point-bias result from the full-affine coverage result. At $p = 500$, even full-affine calibrated SUDO has bias -0.048 and fails the point gate.

With continuous treatment, bias remains below 1.5% of the truth in the binary and ordinal designs. Strong-signal coverage is less stable, reinforcing the need to rerun the full-pipeline bootstrap rather than transfer an uncertainty result from a binary-treatment experiment.

5.1 Count oracle bridge

To separate count completion from fitted-learner error, we supply the true scalar index and count CDF. Six prespecified cells cross Poisson and negative-binomial laws, binary and continuous treatments, and two signal strengths where applicable. At $n = 2000$, with 100 replications, $B = 25$ randomized completions, five folds, and 99 bootstrap resamples, both analytic Rao-Blackwell and randomized completion pass every bias, coverage, and SD-to-SE gate (Table 7). Their paired differences are within two completion Monte Carlo standard errors in every cell.

Table 7: Count oracle bridge with the true index and CDF supplied. Entries pair Rao-Blackwell and randomized completion. Each cell uses $n = 2000$, 100 replications, five folds, and 99 bootstrap resamples; randomized SUDO uses $B = 25$.

Design	θ_0	Bias (RB / randomized)	SD/SE (RB / randomized)	Coverage (RB / randomized)
Continuous, Poisson	$\log(1.5)$	-0.001 / -0.001	0.989 / 0.989	0.970 / 0.970
Continuous, NB	$\log(1.5)$	+0.001 / +0.001	1.046 / 1.042	0.940 / 0.940
Continuous, Poisson	$\log(2)$	+0.001 / +0.001	0.921 / 0.925	0.980 / 0.980
Continuous, NB	$\log(2)$	-0.001 / -0.001	0.963 / 0.962	0.940 / 0.950
Binary, Poisson	$\log(2)$	-0.001 / -0.001	1.127 / 1.120	0.910 / 0.920
Binary, NB	$\log(2)$	-0.002 / -0.002	1.114 / 1.105	0.920 / 0.920

The oracle results validate count completion and bootstrap inference under a correct supplied distribution. They do not validate estimation of that distribution. Constrained XGBoost failed the point-bias screen, so the paper makes no claim for fitted count learners or the planned bikeshare analysis.

6 Wine quality application

We apply ordinal SUDO to the UCI wine-quality data (Cortez et al. 2009). The outcome is the median ordinal score from blind tasters. We analyze 1,599 red and 4,898 white wines separately; scores range from 3 to 8 for red wine and 3 to 9 for white. The exposure is standardized volatile acidity, an acetic-acid fault associated with vinegar-like aroma. The other ten standardized physicochemical measurements form the adjustment set.

This is an observational analysis. A causal interpretation would require the measured chemistry to suffice for adjustment and winemaking practice to affect quality only through that chemistry and volatile acidity (Figure 2). Those assumptions are not established, so we report conditional associations.

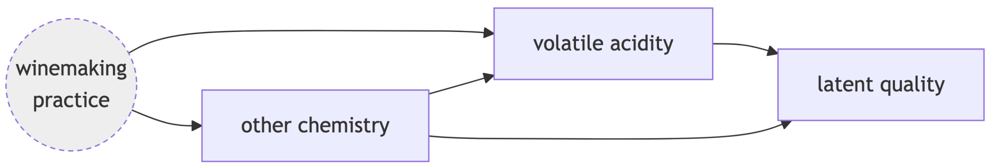

Figure 2: Assumed application DAG. Volatile acidity (the treatment) affects latent quality directly; unobserved winemaking practice confounds both through the measured chemistry, the adjustment set.

Volatile acidity retains 75% of its marginal standard deviation after adjustment in red wine and 94% in white wine. Both imputation models are proportional-odds logits with volatile acidity entering linearly. One uses the adjustment variables linearly; the other uses a three-degree-of-freedom natural spline for each variable, with no treatment-by-covariate interactions. Ordinal SUDO uses 50 proper draws, five-fold cross-fitted GAM nuisances, and Rubin pooling. The two specifications give similar estimates (Table 8).

The coefficients correspond to one standard deviation of volatile acidity, about 0.18 g/dm^3 for red wine and 0.10 for white. A one-SD increase is associated with odds ratios of about 0.59 and 0.64, respectively, for scoring above any fixed quality level. The direct cumulative-link coefficient is a model-based comparator.

Table 8: Effect of a one-SD increase in volatile acidity on latent wine quality. Model-based coefficient from the linear cumulative-link fit (standard error in parentheses); SUDO estimates with 95% intervals in brackets.

Wine	Overlap $R^2(D \sim X)$	Cumulative-link	SUDO (linear)	SUDO (spline)
Red	0.44	-0.61 (0.07)	-0.52 [-0.68, -0.36]	-0.52 [-0.69, -0.35]
White	0.12	-0.50 (0.03)	-0.45 [-0.58, -0.33]	-0.45 [-0.58, -0.32]

The negative sign is stable when any one adjustment covariate is removed. An omitted-variable calibration suggests that a confounder would need to explain roughly 15% of the residual treatment and outcome variation for red wine and 10% for white wine to move the estimate to zero (Cinelli and Hazlett 2020). Because that calibration was derived for linear regression with an observed outcome, we report it as a heuristic rather than a formal SUDO bound.

The fitted pass-through averages 0.21 for red wine and 0.17 for white. Thus the observed categories correct much of a local fitted-index error in these data. This calculation does not establish robustness to arbitrary misspecification.

7 Discussion

SUDO separates two tasks that are often conflated. The fitted distribution places the outcome on the target scale, and DML removes flexible covariate effects on that scale. This gives binary and ordinal models the same partialling-out interface. It does not supersede a correct direct estimator or a tailored orthogonal score. Proper draws or a full-pipeline bootstrap are part of the procedure because a single completion hides uncertainty.

The separation also identifies the weak point. Orthogonality covers the two adjustment regressions, not the distribution that creates the surrogate. Partial linearity rules out treatment-by-covariate interactions but does not ensure a small target bias. Scalar recalibration repairs one important case, namely an index with the right shape and the wrong location or scale. It does not repair arbitrary index error, as the $p = 500$ failure shows.

Current evidence supports binary and ordinal SUDO, including a coefficientless calibrated binary learner. Count completion works with an oracle index and CDF, but fitted count learning remains unresolved. The next inferential check is full coverage validation of scale-only calibration. More broadly, every new imputation learner should be evaluated through target-level bias and coverage experiments.

8 Data and software

The wine-quality data come from the University of California, Irvine Machine Learning Repository (Cortez et al. 2009). The `sudo` Python package provides the parametric estimator for binary and ordinal outcomes, including proper draws and the per-completion ordinal outcome-nuisance refit. The partially-linear machine-learning imputation learners and the full-pipeline bootstrap are implemented in R, which is also where every method is prototyped and validated before release. Count and flexible ordinal adapters remain R-only until their fitted-learner confirmation stages pass. The full-affine calibrated estimator also remains R-only, and scale-only calibration still awaits coverage validation. The open-source repository contains the code and committed result files behind every table and figure under the MIT License: https://github.com/bgreenwell/sudo.

9 Declaration of artificial intelligence use

We used generative artificial intelligence (AI) to assist with code and summaries of reference materials. The authors reviewed and validated all outputs, retain full ownership of the final manuscript, and accept sole responsibility for any errors.

10 Appendix: simulation and application details

Each design draws a latent utility $Z = \theta_0 D + g_0(X) + \varepsilon$ and converts it to an observed category. Most designs use logistic ε; the exceptions are specified below. The designs vary the covariate structure, cutpoints, and fitted models; covariates remain standard normal throughout. The `R/` and `R/results/` directories contain the full data-generating code and committed per-cell results.

10.1 Data-generating processes

All designs instantiate Equation 11 with one of two covariate structures:

- **S1**, one covariate: $p = 1$, $\pi(X) = 0.8X$, $g_0(X) = X$. Every model in play is correctly specified, so only the pooling is under test.
- **S5**, five covariates: $p = 5$, $\pi(X) = X_4 + \cos X_5$, $g_0(X) = X_1^2 + \sin X_2 + 0.5X_3$. The outcome structure is nonlinear in X, and treatment depends on covariates absent from g_0.

Table 9: Data-generating processes. κ lists the interior cutpoints, so (0) is binary and a pair gives $J = 3$.

Design	Structure	F	κ	θ_0	n	Replications
Inference ladder (Table 1)	S1	logistic	(0)	1.5	5000	500
Ordinal parametric (Table 2, top)	S1	logistic	$(-1, 1)$	1.0	3000	500
Binary ML	S5	logistic	(0)	0.5, 3	1000, 2000	200
Ordinal flexible adjustment (Table 2, bottom)	S5	logistic	$(0, 2)$	1.0	1000, 2000	200
Link misspecification	S1	Gumbel	(0) or $(0, 2)$	1.0	5000	200
Black-box imputation	S5	logistic	(0)	1.5, 3	2000	100
PL learner comparison (Table 4)	S5	logistic	(0)	1.5, 3	2000	50

Design	Structure	F	κ	θ_0	n	Replications
Calibration (Table 5)	$p = 80$ additive	logistic	(0)	0.75	3000	50
DGP variation, continuous D	S5, $D = 0.7\pi(X) + N(0, 1)$	logistic	(0) or $(0, 2)$	1, 2.5	2000	200

Models fitted. The ladder uses $\mathrm{glm}(Y \sim D + X)$ as the imputation model with $B \in \{10, 25, 50\}$ draws (the table shows $B = 25$). The ordinal parametric design uses `ordinal::clm`, whose covariance covers the thresholds, with $B = 25$. The ordinal flexible-adjustment design uses parametric `clm` on D and natural-spline bases $\mathrm{ns}(X_k, 4)$ with `mgcv::gam` partialling nuisances and $B = 25$.

Link misspecification. This design retains the one-covariate structure but replaces logistic ε with an extreme-value error: Gumbel-max for the binary case, matching `glm` with a cloglog link, and Gumbel-min for the ordinal case, matching `ordinal::clm` with a cloglog link. The analyst fits either logit (misspecified) or cloglog (correct). Holding the simple covariate structure fixed isolates error in the link.

Partially-linear imputation models. The original black-box result uses $\hat{\eta} = \hat{\alpha} D + \hat{f}(X)$ with $\hat{f}$ a single-hidden-layer neural net (`nnet`, `size = 4`, `decay = 0.3`, five backfitting iterations, selected on target-level Monte Carlo bias) and 100 full-pipeline bootstrap resamples. The common comparison uses 50 resamples per arm. The GAM includes a linear treatment term and smooths of X. MARS builds degree-two bases from X only before a joint logistic fit with mandatory linear D.

Calibration experiment. The calibration designs use $p = 80$ independent standard-normal covariates and

$$g_0(X) = 0.65X_1 + 0.55(X_2^2 - 1) + 0.45X_3 - 0.35(X_4^2 - 1) + 0.30X_5 - 0.25X_6.$$

Treatment is $N(0, 1)$ in the randomized design. In the confounded design, $D = m_0(X) + N(0, 1)$ with $m_0(X) = 0.50X_1 - 0.40X_3 + 0.30(X_2^2 - 1)$. The high-signal design uses $D \sim N(0, 4)$. Outcomes follow $Y \sim \mathrm{Bernoulli}[\Lambda\{0.75D + g_0(X)\}]$. GAMSEL uses cubic candidate smooths for X, a linear treatment term, and cross-validated binomial deviance. Five outer and five inner folds are used. The point ablation uses separate seeds from tuning and coverage.

Wine application. Red and white wines are fit separately. The outcome levels are 3–8 for red wine and 3–9 for white. Volatile acidity is the standardized treatment. The ten standardized adjustment variables are fixed acidity, citric acid, residual sugar, chlorides, free and total sulfur dioxide, density, pH, sulphates, and alcohol. The proportional-odds logit full model is `ordinal::clm` with index $\alpha D + \beta^\top X$ (linear) or $\alpha D + \sum_k \mathrm{ns}(X_k, 3)$ (spline). Partialling nuisances $E[S \mid X]$ and $E[D \mid X]$ use `mgcv::gam` (Gaussian for continuous D), with five-fold cross-fitting, $B = 50$ proper draws, and Rubin pooling.

10.2 Nuisances, cross-fitting, and draws

Partialling nuisances $\hat{\ell}(X) = E[S \mid X]$ and $\hat{m}(X) = E[D \mid X]$ are `mgcv::gam` throughout, fit with five-fold cross-fitting. Proper draws sample the full-model parameters from $N(\hat{\beta}, \widehat{\mathrm{Cov}}(\hat{\beta}))$ (thresholds included for `clm`) before each completion. Diagnostic draws use a random number generator (RNG) stream independent of the estimation draws.

Binary experiments found per-completion refitting of $\hat{\ell}$ negligible, so the binary estimator reuses a plug-in fit. The ordinal estimator refits $\hat{\ell}$ for every completion because reuse inflated its within-draw variance.

10.3 Monte Carlo precision

Coverage is a binomial proportion, so its Monte Carlo standard error at a true 0.95 is about $\sqrt{0.95 \cdot 0.05/R}$: 0.010 at $R = 500$ replications, 0.015 at $R = 200$, and 0.022 at $R = 100$. Per-cell bias and coverage Monte Carlo standard errors are in the committed result files.

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